Geometric Sequences and Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Geometric sequences

Consider the sequence $3, 6, 12, 24, \ldots$

a State the common ratio. $r = 2$ **b** Write a formula for the nth term. $a_n = 3 \cdot 2^{n-1}$ **c** Find a_{10}. $a_{10} = 1536$

(a) $r = 2$; (b) $a_n = 3 \cdot 2^{n-1}$; (c) $a_{10} = 1536$

2 [Foundational] • Target: Geometric sequences

In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.

$r^3 = \frac{48}{6} = 8$, so $r = 2$ and $a_1 = 3$.

3 [Foundational] • Target: Geometric series

Find the sum of the first 8 terms of $5 + 10 + 20 + \cdots$

$S_8 = 5\,\frac{2^8 - 1}{2 - 1} = 5(255) = 1275$.

4 [Foundational] • Target: Geometric series

Evaluate each infinite sum, or state that it diverges:

a $18 + 6 + 2 + \frac{2}{3} + \cdots$ $r = \frac{1}{3}$: $S = \frac{18}{1 - 1/3} = 27$.

b $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$ $r = -\frac{1}{2}$: $S = \frac{1}{1 + 1/2} = \frac{2}{3}$.

c $2 + 3 + 4.5 + \cdots$ $r = 1.5 > 1$: diverges.

(a) $r = \frac{1}{3}$: $S = \frac{18}{1 - 1/3} = 27$.; (b) $r = -\frac{1}{2}$: $S = \frac{1}{1 + 1/2} = \frac{2}{3}$.; (c) $r = 1.5 > 1$: diverges.

5 [Foundational] • Target: Geometric series

Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.

$0.7 + 0.07 + \cdots$ with $a = \frac{7}{10}$, $r = \frac{1}{10}$: $S = \frac{7/10}{9/10} = \frac{7}{9}$.

6 [Foundational] • Target: Geometric series

A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.

$\text{Total} = 8 + 2\left(8 \cdot \frac{3}{4} + 8 \cdot \left(\frac{3}{4}\right)^2 + \cdots\right) = 8 + 2\,\frac{6}{1 - 3/4} = 8 + 48 = 56$ m.

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7 [Foundational] • Target: Geometric means and mixed sequences

Insert two positive geometric means between 4 and 108 (i.e., find b_1, b_2 so that $4, b_1, b_2, 108$ is geometric).

4 terms total: $r^3 = \frac{108}{4} = 27$, so $r = 3$. Means: $12, 36$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric means and mixed sequences

Determine whether each sequence is arithmetic, geometric, or neither:

a $2, 6, 18, 54, \ldots$ Geometric ($r = 3$)

b $5, 8, 11, 14, \ldots$ Arithmetic ($d = 3$)

c $1, 4, 9, 16, \ldots$ Neither (differences $3, 5, 7$ are not constant, and ratios are not constant)

(a) Geometric ($r = 3$); (b) Arithmetic ($d = 3$); (c) Neither (differences $3, 5, 7$ are not constant, and ratios are not constant)

9 [Intermediate] • Target: Applications

A car worth 80,000 riyals depreciates by 15% per year.

a Write a formula for its value after n years. $V_n = 80000(0.85)^n$.

b Find its value after 5 years, to the nearest riyal. $V_5 = 80000(0.85)^5 \approx 80000(0.4437) \approx 35{,}497$ riyals.

(a) $V_n = 80000(0.85)^n$.; (b) $V_5 = 80000(0.85)^5 \approx 80000(0.4437) \approx 35{,}497$ riyals.

10 [Intermediate] • Target: Applications

A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).

This is a geometric series with $a = 1$, $r = 5$, $n = 6$: $S_6 = \frac{1(5^6 - 1)}{5 - 1} = \frac{15624}{4} = 3906$.

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11 [Intermediate] • Target: Visualizing geometric growth

The first 6 terms of $a_n = 2(1.5)^{n-1}$ are plotted below. Contrast the shape of this graph with the straight-line shape of an arithmetic sequence's graph.

Unlike an arithmetic sequence's straight-line pattern, the geometric sequence's points curve upward with increasing steepness — each term is a constant MULTIPLE of the previous one rather than a constant ADDITION, producing exponential-style curvature rather than a line.

12 [Intermediate] • Target: Determining convergence before summing

For each infinite geometric series, determine whether it converges WITHOUT computing the sum, by checking $|r| < 1$: $\sum_{n=1}^{\infty} 5(0.9)^{n-1}$, $\sum_{n=1}^{\infty} 3(1.2)^{n-1}$, $\sum_{n=1}^{\infty} 7(-0.4)^{n-1}$.

First: $r = 0.9$, $|r| < 1$: converges. Second: $r = 1.2$, $|r| > 1$: diverges. Third: $r = -0.4$, $|r| < 1$: converges.

13 [Intermediate] • Target: Finding the number of terms from a partial sum

For the geometric series $2 + 6 + 18 + \cdots$, find the number of terms n needed for the sum to first exceed 2000.

$S_n = \frac{2(3^n - 1)}{3 - 1} = 3^n - 1 > 2000 \Rightarrow 3^n > 2001$. Testing: $3^6 = 729$, $3^7 = 2187$. So $n = 7$ terms are needed (giving $S_7 = 3^7 - 1 = 2186 > 2000$).

14 [Intermediate] • Target: A mixed application: population with geometric decline

A population of an endangered species declines by 12% each year from an initial 5000. Find the total number of 'animal-years' lived by the population over the first 10 years (i.e. the sum of the population sizes at the start of each of the 10 years), to the nearest whole number.

$a_1 = 5000$, $r = 0.88$. $S_{10} = \frac{5000(1 - 0.88^{10})}{1 - 0.88} = \frac{5000(1 - 0.2785)}{0.12} = \frac{5000(0.7215)}{0.12} \approx 30{,}063$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Geometric means and mixed sequences • Cross-Lesson: 4.03 Hyperbolas

Find the sum of the infinite geometric series with $a_1 = 12$ and $r = \frac{1}{3}$.

Final Answer: 18. Applying Sequences principles, $S = \frac{12}{1 - \frac{1}{3}} = 18$. Therefore, the verified correct option is 18.

16 [Challenging] • Target: Applications • Cross-Lesson: 5.01 Vectors: Components and Operations

Find the sum of the infinite geometric series with $a_1 = 20$ and $r = \frac{3}{4}$.

Final Answer: 80. Applying Sequences principles, $S = \frac{20}{1 - \frac{3}{4}} = 80$. Therefore, the verified correct option is 80.

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17 [Challenging] • Target: Visualizing geometric growth • Cross-Lesson: 5.02 The Dot Product

Find the sum of the infinite geometric series with $a_1 = 18$ and $r = \frac{2}{3}$.

Final Answer: 54. Applying Sequences principles, $S = \frac{18}{1 - \frac{2}{3}} = 54$. Therefore, the verified correct option is 54.

18 [Challenging] • Target: Determining convergence before summing • Cross-Lesson: 5.03 Parametric Equations

Find the sum of the infinite geometric series with $a_1 = 8$ and $r = \frac{1}{2}$.

Final Answer: 16. Applying Sequences principles, $S = \frac{8}{1 - \frac{1}{2}} = 16$. Therefore, the verified correct option is 16.

19 [Challenging] • Target: Finding the number of terms from a partial sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the sum of the infinite geometric series with $a_1 = 12$ and $r = -\frac{1}{3}$.

Final Answer: 9. Applying Sequences principles, $S = \frac{12}{1 + \frac{1}{3}} = 9$. Therefore, the verified correct option is 9.

20 [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of the infinite geometric series with $a_1 = 6$ and $r = \frac{1}{4}$.

Final Answer: 8. Applying Sequences principles, $S = \frac{6}{1 - \frac{1}{4}} = 8$. Therefore, the verified correct option is 8.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: General Term of a Geometric Sequence

State the formula for the n-th term a_n of a geometric sequence with first term a_1 and common ratio r.

a. $a_n = a_1 r^{n-1}$ ✓

b. $a_n = a_1 r^n$

c. $a_n = a_1 + (n-1)r$

d. $a_n = \frac{a_1}{r^{n-1}}$

In a geometric sequence, each successive term is multiplied by r, giving $a_n = a_1 r^{n-1}$.

22 [Foundational] • Target: General Term of a Geometric Sequence

Find the 6th term of the geometric sequence $3, 6, 12, 24, \ldots$

a. 96 ✓

b. 192

c. 48

d. 128

$a_1 = 3, r = 2$. $a_6 = 3 \cdot 2^{6-1} = 3 \cdot 2^5 = 3 \cdot 32 = 96$.

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23 [Foundational] • Target: General Term of a Geometric Sequence

In a geometric sequence, $a_2 = 12$ and $a_5 = 324$. Find the common ratio r.

a. 3 ✓ **b.** 2

c. 4 **d.** $\sqrt{3}$

$\frac{a_5}{a_2} = r^3 = \frac{324}{12} = 27 \Rightarrow r = 3$.

24 [Foundational] • Target: Finite Geometric Series

State the formula for the sum S_n of the first n terms of a geometric series ($r \neq 1$).

a. $S_n = \frac{a_1(1-r^n)}{1-r}$ ✓ **b.** $S_n = \frac{a_1}{1-r}$

c. $S_n = \frac{n}{2}(a_1 + a_n)$ **d.** $S_n = a_1(1-r^n)$

The finite sum of a geometric series is $S_n = \frac{a_1(1-r^n)}{1-r}$.

25 [Foundational] • Target: Finite Geometric Series

Find the sum of the first 6 terms of the series $4 + 12 + 36 + \ldots$

a. 1456 ✓ **b.** 1458

c. 486 **d.** 2916

$a_1 = 4, r = 3, n = 6$. $S_6 = \frac{4(1-3^6)}{1-3} = \frac{4(1-729)}{-2} = -2(-728) = 1456$.

26 [Foundational] • Target: Finite Geometric Series

Evaluate $\sum_{k=1}^{5} 2(3)^{k-1}$.

a. 242 ✓ **b.** 162

c. 243 **d.** 486

$a_1 = 2, r = 3, n = 5$. $S_5 = \frac{2(1-3^5)}{1-3} = \frac{2(1-243)}{-2} = 242$.

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27 [Foundational] • Target: Infinite Geometric Series

Under what condition does an infinite geometric series $\sum_{n=1}^{\infty} a_1 r^{n-1}$ converge?

a. $|r| < 1$ ✓

b. $r > 1$

c. $|r| \leq 1$

d. $r > 0$

An infinite geometric series converges if and only if the absolute value of the common ratio is strictly less than 1 ($|r| < 1$).

28 [Foundational] • Target: Infinite Geometric Series

Find the sum of the infinite series $12 + 4 + \frac{4}{3} + \frac{4}{9} + \ldots$

a. 18 ✓

b. 16

c. 24

d. 15

$a_1 = 12, r = 1/3$. Since $|1/3| < 1$, sum $S = \frac{a_1}{1-r} = \frac{12}{1-1/3} = \frac{12}{2/3} = 18$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Infinite Geometric Series

Find the fractional equivalent of the repeating decimal $0.\overline{45} = 0.454545\ldots$ using an infinite geometric series.

a. 5/11 ✓

b. 45/100

c. 9/22

d. 45/90

$a_1 = 45/100, r = 1/100$. $S = \frac{45/100}{1-1/100} = \frac{45/100}{99/100} = \frac{45}{99} = \frac{5}{11}$.

30 [Intermediate] • Target: Geometric Word Problems and Annuities

A bouncing ball is dropped from a height of 16 ft. On each bounce, it rebounds to 3/4 of its previous height. Find the total vertical distance the ball travels before coming to rest.

a. 112 ft ✓

b. 64 ft

c. 128 ft

d. 96 ft

Initial drop $= 16$. Subsequent up-and-down trips form an infinite series with first term $2(16 \times 3/4) = 24$ and ratio 3/4. Total $= 16 + \frac{24}{1-3/4} = 16 + \frac{24}{1/4} = 16 + 96 = 112$ ft.

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31 [Intermediate] • Target: Geometric Word Problems and Annuities

A company offers a starting salary of 50000 dollars with a guaranteed 4% annual raise. What is the salary in the 5th year?

a. $50000(1.04)^4 \approx 58492.93$ dollars ✓

b. 60000 dollars

c. 58000 dollars

d. $50000(1.04)^5$

5th year salary is $a_5 = a_1 r^4 = 50000(1.04)^4 \approx 58492.93$ dollars.

32 [Intermediate] • Target: Geometric Word Problems and Annuities

An ordinary annuity pays 100 dollars at the end of each year for 10 years at 5% annual interest. Which expression represents its future value?

a. $100\left[\frac{(1.05)^{10}-1}{0.05}\right]$ ✓

b. $100(1.05)^{10}$

c. $1000(1.05)$

d. $\frac{100}{0.05}$

The sum of an annuity is a finite geometric series with formula $FV = P\left[\frac{(1+r)^n-1}{r}\right] = 100\left[\frac{(1.05)^{10}-1}{0.05}\right]$.

33 [Intermediate] • Target: Geometric sequences

Find the sum of the infinite geometric series with $a_1 = 4$ and $r = \frac{1}{2}$.

a. 10

b. 4

c. **8** ✓

d. 11

Applying Sequences principles, $S = \frac{4}{1-\frac{1}{2}} = 8$. Therefore, the verified correct option is 8.

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34 [Intermediate] • Target: Geometric series

Find the sum of the first 5 terms of the geometric series with $a_1 = 1$ and $r = \frac{1}{2}$.

a. $\frac{15}{16}$

b. 5

c. $\frac{47}{16}$

d. $\frac{31}{16}$ ✓

$S_n = \frac{a_1(1-r^n)}{1-r} = \frac{1(1-(\frac{1}{2})^5)}{1-(\frac{1}{2})} = \frac{31}{16}$.

35 [Intermediate] • Target: Geometric means and mixed sequences

Find the sum of the first 3 terms of the geometric series with $a_1 = 3$ and $r = \frac{1}{2}$.

a. $\frac{17}{4}$

b. $\frac{25}{4}$

c. 9

d. $\frac{21}{4}$ ✓

$S_n = \frac{a_1(1-r^n)}{1-r} = \frac{3(1-(\frac{1}{2})^3)}{1-(\frac{1}{2})} = \frac{21}{4}$.

36 [Intermediate] • Target: Applications

Find the sum of the infinite geometric series with $a_1 = 5$ and $r = -\frac{1}{3}$.

a. $\frac{15}{4}$ ✓

b. $\frac{11}{4}$

c. 10

d. $\frac{19}{4}$

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{5}{1-(-\frac{1}{3})} = \frac{15}{4}$.

37 [Intermediate] • Target: Visualizing geometric growth

Find the sum of the infinite geometric series with $a_1 = 9$ and $r = -\frac{1}{3}$.

a. $\frac{23}{4}$

b. $\frac{27}{4}$ ✓

c. 18

d. $\frac{31}{4}$

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{9}{1-(-\frac{1}{3})} = \frac{27}{4}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Determining convergence before summing • Cross-Lesson: 5.03 Parametric Equations

Find the sum of the infinite geometric series with $a_1 = 8$ and $r = -\frac{1}{2}$.

a. $\frac{19}{3}$

b. $\frac{13}{3}$

c. 16

d. $\frac{16}{3}$ ✓

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{8}{1-(-\frac{1}{2})} = \frac{16}{3}$.

39 [Challenging] • Target: Finding the number of terms from a partial sum • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the sum of the infinite geometric series with $a_1 = 6$ and $r = -\frac{1}{3}$.

a. $\frac{11}{2}$

b. $\frac{7}{2}$

c. 12

d. $\frac{9}{2}$ ✓

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{6}{1-(-\frac{1}{3})} = \frac{9}{2}$.

40 [Challenging] • Target: A mixed application: population with geometric decline • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of the infinite geometric series with $a_1 = 8$ and $r = \frac{1}{4}$.

a. $\frac{29}{3}$

b. $\frac{32}{3}$ ✓

c. $\frac{35}{3}$

d. 16

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{8}{1-(\frac{1}{4})} = \frac{32}{3}$.

41 [Challenging] • Target: General Term of a Geometric Sequence • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the sum of the infinite geometric series with $a_1 = 3$ and $r = -\frac{1}{3}$.

a. $\frac{9}{4}$ ✓

b. 6

c. $\frac{13}{4}$

d. $\frac{5}{4}$

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{3}{1-(-\frac{1}{3})} = \frac{9}{4}$.

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42 [Challenging] • Target: Finite Geometric Series • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the sum of the infinite geometric series with $a_1 = 5$ and $r = \frac{1}{4}$.

a. 10

b. $\frac{23}{3}$

c. $\frac{17}{3}$

d. $\frac{20}{3}$ ✓

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{5}{1-(\frac{1}{4})} = \frac{20}{3}$.

43 [Challenging] • Target: Infinite Geometric Series • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the sum of the infinite geometric series with $a_1 = 4$ and $r = -\frac{1}{2}$.

a. $\frac{5}{3}$

b. 8

c. $\frac{11}{3}$

d. $\frac{8}{3}$ ✓

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{4}{1-(-\frac{1}{2})} = \frac{8}{3}$.

44 [Challenging] • Target: Geometric Word Problems and Annuities • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the sum of the infinite geometric series with $a_1 = 2$ and $r = -\frac{1}{3}$.

a. $\frac{1}{2}$

b. $\frac{5}{2}$

c. 4

d. $\frac{3}{2}$ ✓

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{2}{1-(-\frac{1}{3})} = \frac{3}{2}$.

45 [Challenging] • Target: Geometric sequences • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the sum of the infinite geometric series with $a_1 = 3$ and $r = \frac{1}{3}$.

a. $\frac{11}{2}$

b. $\frac{9}{2}$ ✓

c. 6

d. $\frac{7}{2}$

Since $|r| < 1$, $S = \frac{a_1}{1-r} = \frac{3}{1-(\frac{1}{3})} = \frac{9}{2}$.

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